A Trigger-Guided FPGA Digital CFD Without an Additional CFD Validation Threshold for High-Efficiency HPGe Detector Timing

H. Y. Wu, Z. H. Li, D. W. Luo, H. Hua, Z. H. Wang, and M. Venaruzzo

***Abstract*—Precise timing combined with high low-energy detection efficiency in high-purity germanium (HPGe) detectors is increasingly important for spectroscopic applications, but remains a significant challenge for digital data acquisition systems. Standard Digital constant fraction discriminators (CFDs) effectively correct the charge collection time walk but typically enforce a hard validation threshold to suppress noise, resulting in a blind spot for low-amplitude events. To address this limitation, we present a trigger-guided digital CFD architecture implemented on the CAEN 27xx Open FPGA platform, in which event detection is performed by a fast trapezoidal trigger while the CFD branch extracts the zero-crossing time without applying an additional CFD validation threshold. Experimental validation using a ^{60}Co source yields a coincidence resolving time (CRT) of 10.65(2) ns, which is comparable to that of state-of-the-art commercial reference systems. The proposed method preserves substantially more timing-accepted events in the low-energy region by avoiding the additional online timing-validation loss observed in the reference processing chain. This method offers a robust solution for high-granularity arrays and gamma-gamma coincidence experiments involving low-energy transitions.**

***Index Terms*—Constant fraction discriminator (CFD), digital pulse processing (DPP), FPGA, high-purity germanium (HPGe).**

I. Introduction

GAMMA spectroscopy arrays consisting of high-purity germanium (HPGe) detectors represent one of the most powerful devices for exploring nuclear structure. The HPGe detectors possess unrivaled energy resolution, and their precise energy determination has been achieved by digital pulse processing (DPP) algorithms, primarily related to the optimization of trapezoidal pulse shapes [1], [2], [3], [4], [5]. Meanwhile, in modern nuclear physics experiments like gamma-gamma coincidence measurements with high-granularity arrays, or neutron inelastic scattering cross section measurements at white neutron sources, the timing performance of HPGe detectors is becoming increasingly critical [6], [7].

Extracting precise timing from HPGe signals is inherently difficult due to the physics of charge collection. Unlike fast scintillators, HPGe pulses exhibit variable rise times (ranging from 50 ns to several hundred ns) depending on the interaction position within the crystal volume [4], [8]. Therefore, ensuring precise time resolution to effectively gate coincidence matrices or reject background is a considerable challenge.

Standard DPP-PHA firmware (e.g., CAEN 274x series) typically utilizes a triangular filter for pulse detection, with the timing derived from the first zero-crossing of the filtered signal. Although this approach enhances the signal-to-noise ratio (SNR) compared to simple threshold triggers, it remains highly sensitive to the variable charge collection times inherent in HPGe detectors. The peak and subsequent zero-crossing of a filtered pulse depend on the input signal's shape, thus events with identical energy but different interaction depths exhibit significant rise time walk, severely degrading the coincidence resolving time (CRT) in large-volume crystals.

The constant fraction discriminator (CFD) is theoretically capable of correcting amplitude walk. Early digital implementations [7] demonstrated that separating the trigger logic from the zero-crossing extraction could improve efficiency. However, these approaches were largely restricted to offline software processing of full waveforms, which requires significant data bandwidth and precludes real-time coincidence gating.

In commercial systems like the XIA Pixie-16 [9], [10], a CFD validation threshold is strictly enforced to suppress baseline noise to enable real-time operation while minimizing false triggers. For HPGe detectors, where low-energy efficiency is often a key metric, this thresholding creates a blind spot: valid low-energy events that successfully trigger the fast energy filter are often rejected by the CFD logic, leading to a loss of timing information for a significant portion of the spectrum.

To overcome these challenges specifically for HPGe applications, a fully pipelined timing strategy on the CAEN Open FPGA platform is implemented. By replacing the differentiation-based trigger often used in offline algorithms with a Fast Trapezoidal Filter, we achieve superior noise immunity, allowing the system to trigger reliably on minimal energy depositions. We propose a trigger-guided digital CFD

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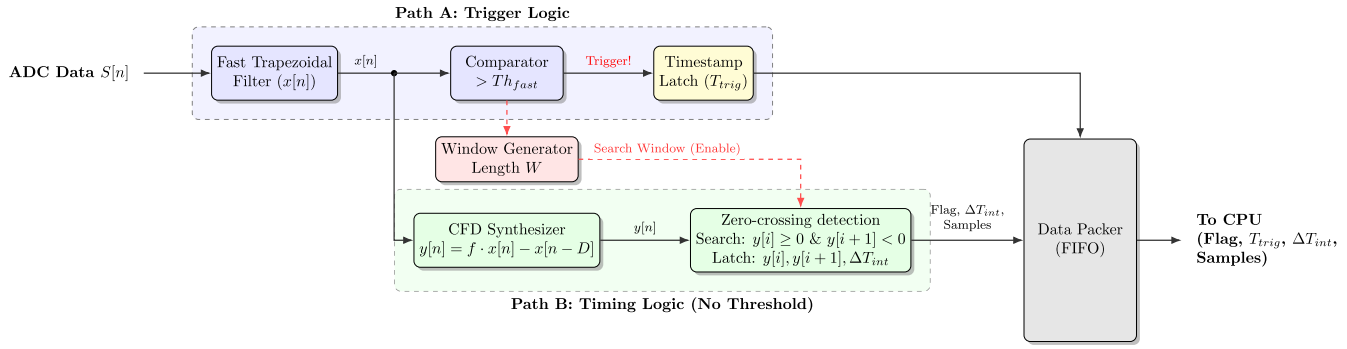


Fig. 1. Simplified logic block diagram of the proposed trigger-guided and digital CFD architecture without an additional CFD validation threshold implemented on the FPGA. The design features two parallel paths: Path A (trigger logic) utilizes a fast trapezoidal filter (x[n]) for high-efficiency event detection, while Path B (timing logic) performs zero-crossing detection on the synthesized CFD signal (y[n]) without an amplitude threshold, gated by the search window from Path A.

without a CFD validation threshold architecture. Key innovations include the following.

1) *Decoupled Logic:* Event detection is governed exclusively by a fast trapezoidal filter optimized for the HPGe preamplifier decay time, enabling robust triggering of low-amplitude signals down to the practical noise limit.
2) *Zero-Crossing Without an Additional CFD Validation Threshold:* Once triggered, the algorithm searches for a zero-crossing in the synthesized CFD signal within a dynamically calculated window. By removing the amplitude threshold on the CFD branch, we recover timing information for low-energy X-rays and γ-rays that would otherwise be discarded by conventional firmware.
3) *Hybrid Data Output:* The system outputs the fast trigger timestamp, the CFD time difference, and the raw CFD signal samples around the zero-crossing point.

This approach is particularly advantageous for HPGe detectors. The preservation of raw zero-crossing samples allows for advanced offline processing. Pulse shape analysis (PSA) techniques can be applied to further correct for the position-dependent rise time walk, which is a leading cause of timing uncertainty in large-volume germanium crystals. It has been demonstrated that this firmware implementation substantially mitigates the low-energy timing-acceptance loss observed in the reference digital CFD implementation while providing excellent time resolution, thereby enhancing the capabilities of HPGe arrays in demanding coincidence experiments.

In gamma-gamma coincidence measurements with HPGe arrays, fine timing precision is important not because the measurement itself has an exceptionally high counting rate, but because narrow prompt time gates are required to suppress accidental coincidences and to establish clean coincidence relationships, especially for weak transitions in complex level schemes. In this sense, timing quality directly affects the peak-to-background ratio and the fidelity of coincidence analysis. This motivates the development of a timing architecture that preserves efficiency while maintaining nanosecond-level timing precision.

II. Algorithm and FPGA Implementation

The proposed DPP architecture is implemented on the CAEN Open FPGA platform (e.g., 27xx). The logic is designed to process the digitized input stream $S[n]$ in real-time. The processing chain consists of three parallel stages: energy filtering [5], fast trigger generation [5], and the novel trigger-guided CFD timing extraction. Unlike standard commercial firmware, where the valid timing signal depends on the CFD amplitude, our architecture strictly separates the event trigger decision from the timing extraction process. The simplified logic block diagram of this architecture is shown in Fig. 1, which highlights the parallel execution of the trigger logic (Path A) and the zero-crossing detection logic (Path B).

A. Fast Trigger Logic

To ensure high detection efficiency for low-energy events, the trigger is derived solely from a fast trapezoidal filter. Unlike differentiation-based trigger methods ($S[n] - S[n-k]$), which inherently amplify high-frequency noise, the trapezoidal filter functions as a weighted moving average. This integration process significantly improves the SNR before the trigger decision, enabling the detection of events that are virtually indistinguishable from the baseline in the raw or differentiated signal domain. The recursive algorithm for the fast filter $x[n]$ is defined as follows:

$$x[n] = x[n-1] + (S[n] - S[n-k]) - (S[n-l] - S[n-l-k]) \quad (1)$$

where k and l represent the peaking time and flat-top parameters, respectively. For HPGe signals, these parameters are tuned to be short (e.g., $l \approx 100$ ns and $k \approx 100$ ns) to maximize pile-up rejection and timing precision while maintaining a sufficient SNR.

An event is declared valid at clock cycle T_{trig} when $x[n]$ crosses a user-defined threshold Th_{fast}. This T_{trig} serves as the coarse timestamp and the reference point for the subsequent CFD search window.

B. CFD Logic Without an Additional Validation Threshold

The CFD signal $y[n]$ is synthesized in parallel with the trigger logic. To accommodate the slow and variable rise times of HPGe detectors, the algorithm applies a programmable delay D and an attenuation fraction f

$$y[n] = f \cdot x[n] - x[n-D] \quad (2)$$

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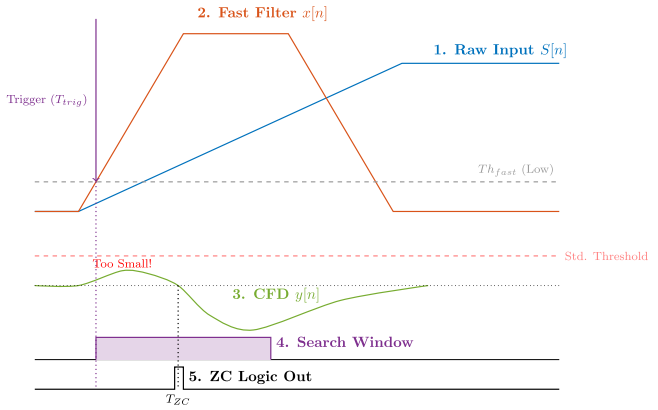


Fig. 2. Conceptual timing diagram illustrating the processing of a low-amplitude HPGe signal.

where $x[n]$ is the output of the fast timing filter described in Section II-A. In traditional architectures, the condition $y[n] > Th_{CFD}$ is required for a validation logic. As discussed, for small-energy deposition in HPGe, the amplitude of $y[n]$ becomes comparable to the baseline noise, causing the traditional logic to fail. A conceptual timing diagram of the proposed method is presented in Fig. 2. It illustrates how a valid zero-crossing from a low-amplitude signal is successfully latched within the trigger-guided search window, though its peak amplitude falls below the standard validation threshold.

In this work, the term "without a CFD validation threshold" refers specifically to the removal of the additional amplitude criterion traditionally applied to the digital CFD branch after event triggering. The system still uses a trigger threshold in the fast-filter branch to declare a valid event.

Instead of checking amplitude, our logic initiates a search window W of length L clocks immediately upon the generation of T_{trig}. The start of the window is offset by the CFD delay parameter to align with the expected zero-crossing region. Inside the window, the logic monitors the sign bit of $y[n]$ to identify a zero-crossing. A valid zero-crossing is defined by a transition from positive to negative

$$y[i] \geq 0 \quad \text{and} \quad y[i+1] < 0.$$

To avoid triggering on high-frequency noise, a slope check can be optionally enabled, requiring $(y[i] - y[i+1]) > S_{min}$.

C. Sub-Sample Timing and Data Output

To minimize FPGA resource utilization and avoid the latency associated with hardware-based division algorithms, the firmware is designed to offload the computational burden of sub-sample interpolation to the offline post-processing stage. Upon identifying the zero-crossing index i within the search window (where $y[i] \geq 0$ and $y[i+1] < 0$), the firmware executes the following operations.

1) *Coarse Latency Calculation (Integer Precision):* Instead of calculating the fractional time on-chip, the FPGA computes the integer time difference, ΔT_{int}, measured in sampling clock cycles. This value represents the temporal distance between the fast trigger timestamp (T_{trig}) and the detected zero-crossing index i

$$\Delta T_{int} = \left(T_{window_start} - T_{trig}\right) + i \tag{3}$$

where T_{window_start} is the timestamp when the search window is opened. This operation requires only simple integer addition logic.
2) *Raw Sample Latching:* Simultaneously, the firmware latches the two raw CFD filter values adjacent to the zero-crossing, $y[i]$ (the sample before or at zero) and $y[i+1]$ (the sample immediately after zero). These signed integer values are preserved to allow for precise reconstruction of the zero-crossing slope downstream. If needed, four or six raw CFD filter samples in the vicinity of the zero-crossing can be retained for further offline analysis, such as refined interpolation or time-walk correction.
3) *Offline Sub-Sample Refinement:* The final picosecond-precision timing is derived during offline analysis (e.g., on a CPU). The precise time difference ΔT_{fine} is calculated using the exported integer delay and the raw samples. Here, T_{clk} denotes the time interval between the two adjacent samples used for the interpolation (i.e., the effective sampling period for the interpolated points)

$$\Delta T_{fine} = \left(\Delta T_{int} + \frac{y[i]}{y[i] - y[i+1]}\right) \times T_{clk}. \tag{4}$$

This division-free hardware approach significantly reduces the FPGA logic footprint (avoiding complex divider IP cores) and accommodates the implementation of more advanced interpolation algorithms (e.g., cubic splines) offline if linear interpolation proves insufficient. Unless otherwise stated, all timing results shown in this work were obtained using only linear interpolation between the two adjacent samples, $y[i]$ and $y[i+1]$, that bracket the timing pickoff point. No energy-dependent time-walk correction, quadratic correction, or offline timing calibration was applied in generating these figures. Therefore, the timing behavior presented in this work reflects the direct performance of the proposed reconstruction method itself, without any additional post-processing compensation.
4) *Data Packet Structure:* For each valid event, the FPGA generates a data packet containing trigger timestamp tagged as T_{trig}, CFD flag with 1-bit valid flag, integer delay tagged as ΔT_{int} (the clock count from trigger to zero-crossing) and diagnostic samples (the raw values $y[i]$, $y[i+1]$ or more). If no zero-crossing is found, the flag is set to 0, and ΔT_{int} and samples are set to null values.

Full-waveform transfer to the host and offline time-walk correction are certainly possible, especially in low-rate studies or dedicated calibration runs. However, for sustained operation in multichannel HPGe arrays, streaming complete raw ADC waveforms for every event significantly increases data throughput, storage burden, and host-side processing latency. The present FPGA implementation, therefore, performs the first-stage timing extraction online and outputs compact observables, including the fast-trigger timestamp, the CFD time difference, and a short segment of raw CFD samples near

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the zero-crossing point. This hybrid strategy preserves the possibility of refined offline correction while maintaining practical online operation.

D. FPGA Implementation Metrics and Data Packet Cost

To facilitate reproducibility and hardware scaling in multichannel HPGe arrays, we summarize the implementation cost of the proposed Path B timing module here. The design was compiled for the Xilinx XCZU19EG-FFVC1760-2LV-E device and operates at a system clock frequency of 125 MHz. For a single-channel implementation, the compiled resource usage is 157 LUTs, 64 LUTRAMs, 206 FFs, 1 DSP block, and no BRAM consumption. These numbers indicate that the proposed trigger-guided CFD branch has a compact hardware footprint and can be replicated across multiple channels with modest resource overhead.

The timing pipeline is fully synchronous with the 125 MHz clock. In the current implementation, the CFD extraction path is realized with a short register-based pipeline, and the latency from trigger assertion to valid timing output is determined by the trigger alignment, delayed-sample access, and zero-crossing decision logic. In practical operation, the module remains limited by the trigger interval and the search window length rather than by the combinational logic itself. Therefore, the maximum sustainable event rate depends on the event spacing and readout bandwidth of the acquisition system; for the present implementation, the timing logic is suitable for continuous operation at the 125 MHz clock domain without introducing a computational bottleneck.

The exported CFD timing packet is intentionally compact. The CFD-valid flag occupies 1 bit, ΔT_{int} is encoded with 8 bits, and each CFD sample used for offline refinement is stored with 24 bits. Therefore, recording two CFD samples requires 57 bits in total. When additional samples are retained, the packet size increases by 48 bits for every extra pair of CFD samples. Specifically, the overhead for saving four and six CFD samples is 105 and 153 bits, respectively. This compact packet format preserves the raw information needed for offline sub-sample interpolation while avoiding the throughput penalty associated with full waveform readout.

III. Experimental Setup

A. Detectors and Signal Chain

The experimental validation was performed using two high-efficiency HPGe detectors. Both detectors are coaxial N-type HPGe crystals with a relative efficiency of 70% (relative to a 3×3 in NaI(Tl) crystal). The detectors were positioned in a close face-to-face geometry (approximately 18 cm apart) with a standard ^{60}Co calibration source placed centrally between them. This setup facilitates gamma-gamma coincidence measurements of the 1173 and 1332 keV cascade, which serves as the benchmark for evaluating timing performance.

The HPGe detector preamplifier features dual identical signal outputs. These outputs were fed independently into the two parallel data acquisition systems [5], [11], ensuring that both the proposed algorithm and the reference commercial system processed identical analog events simultaneously, without the signal attenuation associated with passive splitters.

For the ^{60}Co timing benchmark, the measured singles counting rates were approximately 2900 counts/s in each detector, and the prompt coincidence rate was about 80 counts/s under the acquisition conditions used in this work.

B. Data Acquisition Systems

The novel timing algorithm described in Section II was implemented on a CAEN 2745 Digitizer, part of the new 27xx Open FPGA series. The CAEN 2745 features a 16-bit ADC with a sampling rate of 125 MS/s and is equipped with a Xilinx Zynq UltraScale+ (XCZU19EG) FPGA, open to user customization via the firmware development kit (FDK). In addition to the timing logic, we integrated a five-segment summation energy filter [5]. Furthermore, these algorithms were also successfully deployed on the higher-speed CAEN 2730 (14-bit, 500 MS/s) and 2751 (14-bit, 1 GS/s) digitizers.

The custom software was configured with a fast trapezoidal filter (rise time = 96 ns, flat top = 96 ns) for triggering. The digital CFD parameters were optimized for HPGe signals with a delay $D = 40$ ns and a fraction $f = 0.25$. The zero-crossing search window W was set to 400 ns to accommodate the variable charge-collection times of the large-volume crystals. The event-detection trigger threshold of the V2745 system was carefully adjusted to correspond to approximately 20 keV.

For comparative analysis, an XIA Pixie-16 (100 MS/s, 14-bit) digitizer was used as the industry-standard [11], [12]. The Pixie-16 firmware implements the standard digital CFD algorithm with an additional CFD validation threshold. In the commercial/firmware settings, this threshold corresponds to the parameter `CFDThresh`, which was set to 10 in the present comparison. This value was determined empirically by progressively lowering the parameter until no further increase was observed in the number of accepted low-energy timing events; the minimum value satisfying this condition was then adopted as the optimal setting.

To ensure a fair comparison, the energy filter settings were matched identically to those of the V2745, with a rise time of 5.04 μs and a flat top of 1.2 μs. Regarding the timing logic, the Pixie-16 fast trapezoidal filter was configured with a 100 ns rise time and a 100 ns flat top, while the CFD parameters were set to their recommended values for large HPGe detectors ($D = 50$ ns, $f = 25\%$). The event-detection trigger threshold of the Pixie-16 was also carefully tuned to correspond to approximately 20 keV. The consistency of the trigger thresholds in the two systems was verified by comparing the low-energy onset in the measured energy spectra and the corresponding count rates, thereby ensuring equivalent triggering conditions for the comparison.

IV. Performance Evaluation

Fig. 3 shows raw ADC pulse shapes selected from the same 662 keV γ-ray. Although the pulse amplitudes are nearly identical, their leading-edge rise times vary significantly, reflecting different interaction locations and charge-collection paths in the HPGe crystal. This rise time dispersion is a major source of time walk in large-volume HPGe detectors and motivates the need for a trigger-guided timing architecture as well as the preservation of local timing samples for further offline correction.

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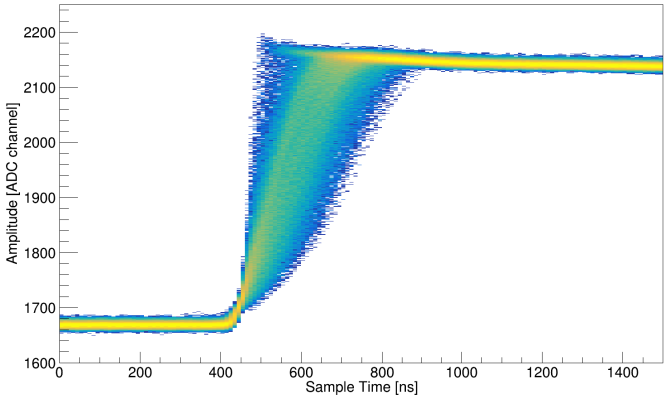


Fig. 3. Raw ADC pulse shapes selected from a ^{137}Cs measurement within a narrow 662 keV energy gate, illustrating rise time variations associated with different interaction points in the Ge crystal.

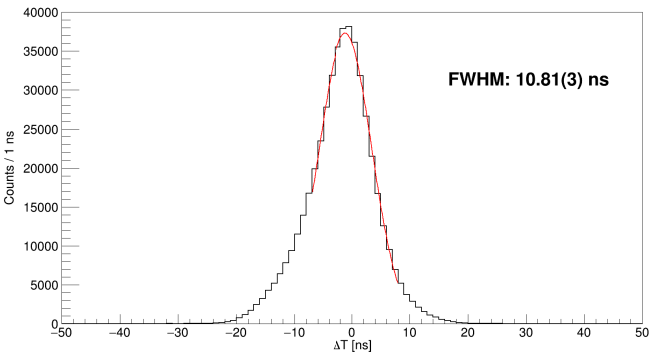


Fig. 5. Coincidence timing spectrum for the ^{60}Co 1173–1332 keV cascade measured with the reference system. The distribution shows a time resolution of 10.81 (3) ns FWHM, establishing a benchmark for timing precision in the high-energy region.

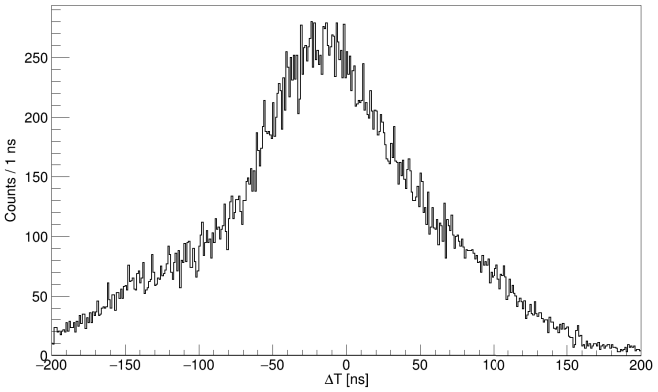


Fig. 4. Coincidence timing spectrum for ^{60}Co measured using the standard CAEN DPP-PHA firmware. The broad distribution (spanning > 100 ns) highlights the severe impact of rise time walk in HPGe detectors, which simple timing algorithms cannot correct.

To rigorously validate the efficacy of the proposed trigger-guided digital CFD without a CFD validation threshold architecture, a comparative study was conducted using a ^{60}Co source for high-precision timing benchmark analysis. The performance of the CAEN 2745 digitizer running the proposed Open FPGA firmware was evaluated against two established industry baselines: 1) the standard CAEN DPP-PHA firmware, representing triangular filter discrimination, and 2) the XIA Pixie-16 system, representing the state-of-the-art digital CFD.

The limitations of the standard timing logic were first established to motivate the necessity of advanced digital processing. As illustrated in Fig. 4, the coincidence timing spectrum for the ^{60}Co 1173–1332 keV cascade measured with the standard CAEN DPP-PHA firmware exhibits a prohibitively broad distribution with a FWHM exceeding 100 ns. This poor performance is the direct consequence of the triangular-filter-based zero-crossing detection used in the standard logic. In large-volume HPGe detectors, interaction-depth variations lead to a wide range of signal rise times. Since the triangular filter's zero-crossing point shifts in response to the pulse shape, this rise time walk is not adequately compensated, making the standard firmware unsuitable for precision coincidence measurements. To address these shortcomings, a new timing architecture is currently under development for future releases of the PHA firmware. This implementation utilizes a pole-zero compensation stage to reduce the time constant of the input signal, followed by a classic digital CFD algorithm applied to the resulting fast exponential pulse. Importantly, this new method is designed as an alternative option rather than a replacement for the standard triangular filter. Both algorithms will be available to the user to accommodate different experimental priorities. For applications focused solely on energy spectroscopy, the triangular filter remains the preferred choice to achieve the best energy resolution and lower energy thresholds. In contrast, the trigger-guided logic presented in this work decouples event detection from timing extraction to recover valid information for low-amplitude transitions that would otherwise be rejected by conventional validation logic.

To address the walk effect, the commercial Pixie-16 system employing a digital CFD algorithm was evaluated. As shown in Fig. 5, this system successfully compensates for the rise time variations, obtaining a sharp CRT of 10.81 (3) ns for the same ^{60}Co cascade. However, this precision comes at the cost of detection efficiency in the low-energy regime. The 2-D correlation matrix of energy versus time difference (ΔT) for the Pixie-16, shown in Fig. 6, reveals a notable sparsity of valid timing events in the low-energy region (<200 keV). It should be noted that Figs. 5 and 6 were obtained from the Pixie-16 timing system rather than from the FPGA timing algorithm proposed in this work. The right-hand tail visible in the full 2-D distribution mainly reflects low-energy time walk, whereas the slight left-hand tail in the gated 1173–1332 keV projection is attributed to residual peak-window selection effects and small channel-dependent timing asymmetries.

To determine more directly whether this low-energy deficit originates from the online timing-validation stage, we performed the same-chain ON/OFF control experiment using the Pixie-16 reference system itself. We performed the same-chain control experiment using the Pixie-16 reference system itself under the same 1332-keV coincidence condition as that used in Fig. 6. In the standard mode, the coincidence spectrum was constructed from the online Pixie-16 timestamps with the firmware CFD timing-validation criterion enabled. In the control mode, this online CFD timing-validation requirement was disabled, the triggered waveforms were recorded, and the event times were extracted offline from the stored traces.

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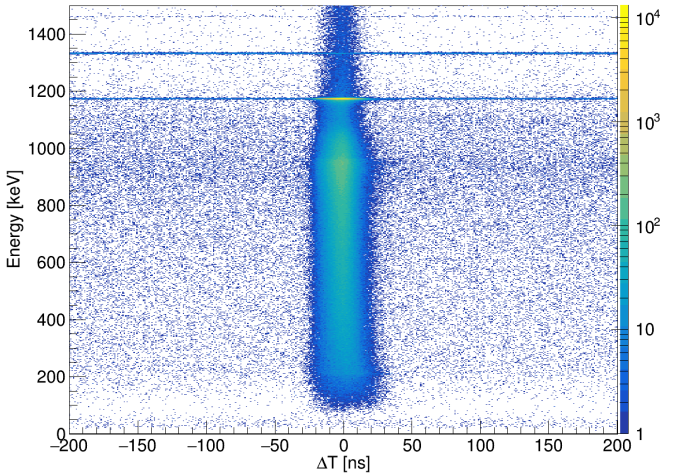


Fig. 6. 2-D matrix from the reference Pixie-16 system. Note the reduced event density (sparsity) in the low-energy region compared to Fig. 9, caused by the CFD threshold cut.

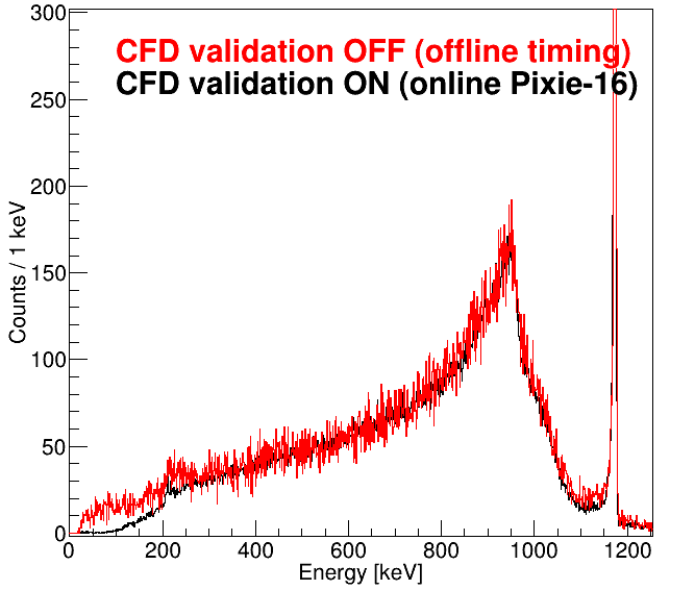


Fig. 7. Comparison of coincidence spectra obtained with the same Pixie-16 system under two timing-processing modes: CFD validation OFF with waveform recording and offline timing extraction using the same CFD parameters and the same linear zero-crossing interpolation as in the online CFD branch (red), and CFD validation ON in the standard online Pixie-16 operation (black). Both spectra were constructed under the same 1332-keV coincidence condition. The recovery of low-energy events in the offline-timed spectrum indicates that the low-energy deficit in the standard reference result is mainly associated with the online CFD timing-validation step.

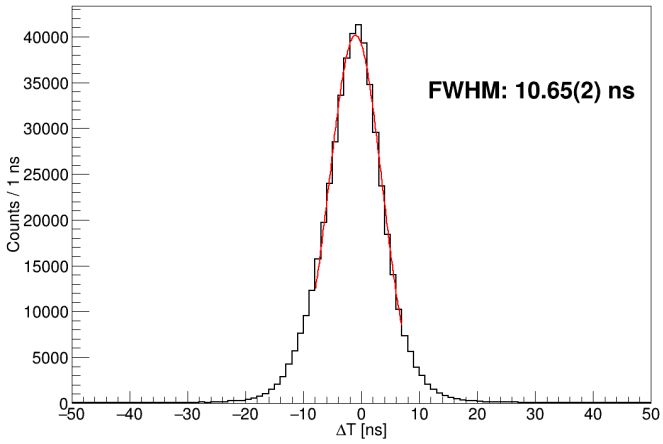


Fig. 8. Coincidence timing spectrum of ^{60}Co (1173–1332 keV) measured with the proposed algorithm. The FWHM is 10.65 (2) ns, matching the precision of the commercial CFD benchmark.

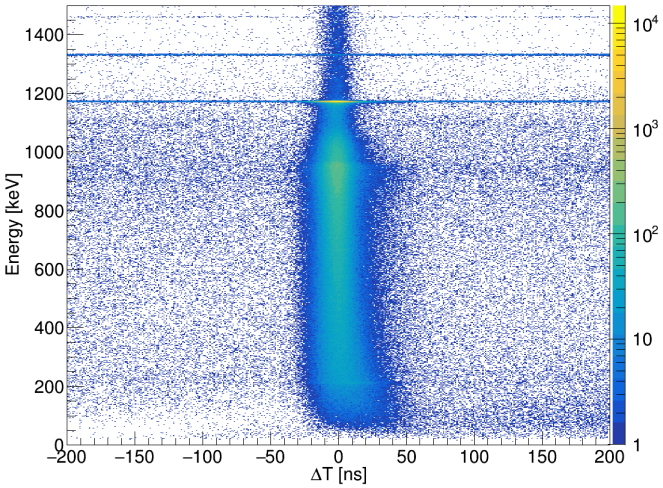


Fig. 9. 2-D histogram of time difference versus energy measured with the proposed algorithm (reference detector gated at 1332 keV). The vertical alignment demonstrates successful timing extraction across the investigated energy range, with residual low-energy walk remaining uncorrected.

As shown in Fig. 7, disabling the online CFD timing-validation step leads to a pronounced recovery of low-energy coincidence events, particularly below 200 keV. The spectra were normalized to the same high-energy peak area. Since these recovered events were recorded as triggered waveforms by the same Pixie-16 system, they demonstrably passed the front-end trigger and were not lost because of an insufficient trigger threshold. Because the comparison is performed within the same digitizer hardware, the observed recovery also cannot be attributed to differences in ADC resolution, sampling rate, or digitizer family. The low-energy sparsity in the standard reference result therefore arises primarily from the online CFD timing-validation stage rather than from the front-end trigger stage.

In contrast, the proposed trigger-guided algorithm implemented on the CAEN 2745 demonstrates the ability to overcome this efficiency-precision trade-off. Fig. 8 presents the coincidence timing spectrum for ^{60}Co obtained with the new firmware. The resulting distribution yields a CRT ofThis result is statistically comparable 10.65 (2) ns. This result is statistically comparable to the 10.81 ns benchmark set by the commercial CFD. It serves as strong evidence that removing the strict validation threshold from the CFD logic does not compromise timing precision, provided that the zero-crossing is correctly isolated within the trigger-guided window.

The core advantage of the proposed architecture, specifically the recovery of low-energy efficiency, is visualized in the 2-D energy-time matrix shown in Fig. 9. The residual curvature visible in Fig. 9 is uncorrected and arises from the raw timestamps reconstructed by this linear-interpolation

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procedure alone. Unlike the reference system (see Fig. 6), which shows a truncated distribution, the proposed algorithm maintains a continuous timing band extending from the 1332 keV photopeak down to the lowest-energy region analyzed (approximately 20 keV). The vertical alignment of this band confirms that the trigger-guided logic successfully identifies the correct zero-crossing point even for events that would fail a standard amplitude threshold check. This result indicates that valid timing information exists and can be extracted for these low-energy events, so long as the trigger source (Path A) is sufficiently sensitive.

To quantify this recovery of detection efficiency, the 2-D datasets from both systems were projected onto the energy axis. A prompt coincidence window of ± 50 ns was applied for the coincidence analysis. This choice is consistent with the measured timing resolution and is sufficiently wide to retain all prompt events of interest, essentially, while still suppressing accidental coincidences effectively. Since the accidental coincidence background scales approximately with the coincidence window width, enlarging the gate from ± 50 to ± 200 ns would increase the accidental contribution by about a factor of four, while providing only marginal gain in true-coincidence efficiency for this system. Therefore, a substantially wider gate would mainly degrade the peak-to-background ratio of the coincidence spectrum rather than improve the useful event yield. The resulting overlaid energy spectra are presented in Fig. 10, which reveals a stark divergence in the low-energy regime between the Pixie-16 and the proposed algorithm. The spectra from the Pixie-16 (black line) and the proposed algorithm (red line) agree well in the high-energy region, as indicated by their alignment at the 1173 and 1332 keV peaks. However, a significant deficit below approximately 400 keV is observed in the reference spectrum obtained with the Pixie-16 system. The proposed method without an additional CFD validation threshold clearly recovers a substantial number of counts in the continuum and low-energy backscatter peaks. This difference directly corresponds to the valid physics events that successfully triggered the energy filter but were subsequently rejected by the commercial CFD's amplitude threshold, thereby supporting the conclusion that the proposed method substantially mitigates the low-energy timing-acceptance loss observed in the commercial CFD processing chain.

While Fig. 10 provides a direct visual comparison between the proposed method and the Pixie-16 reference system, a more quantitative evaluation is given in Fig. 11. In this analysis, the accepted timing-event counts from the two methods were compared at several representative energies using the same acquisition live time. For each energy point, the counts were integrated within an interval of $E \pm 20$ keV, and the ratio between the proposed method and the Pixie-16 system was calculated.

For all points, the same coincidence time gate as that used in Figs. 6, 8, and 9 were applied to both datasets, and the spectra were analyzed using the same histogram binning. No additional energy-dependent normalization was introduced; the two datasets were normalized only by acquisition live time. The vertical error bars were obtained by propagating Poisson counting uncertainties of the integrated counts in the two spectra.

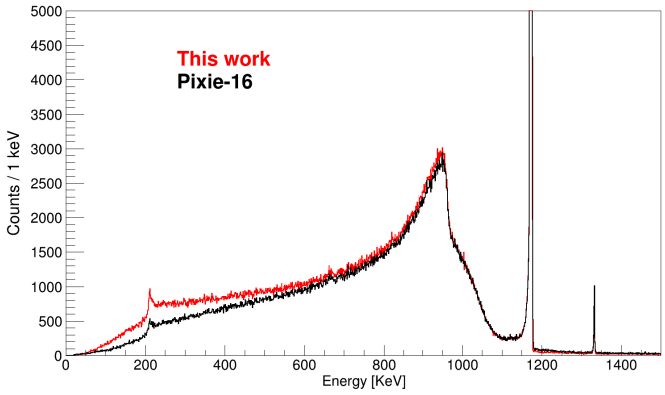


Fig. 10. Comparison of coincident energy spectra obtained by projecting the energy-time matrices (Figs. 6 and 9) onto the energy axis with a time gate of [−50, 50] ns. The red line (this work) shows significantly higher efficiency in the low-energy region (<400 keV) compared to the black line (Pixie-16), demonstrating the recovery of events that fall below the commercial CFD's validation threshold.

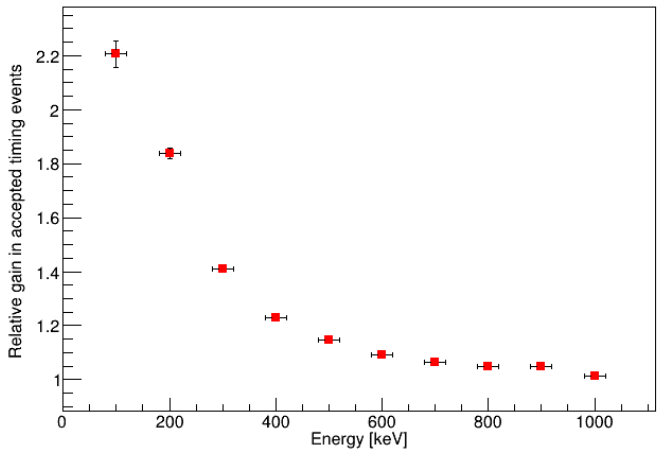


Fig. 11. Energy dependence of the relative timing-accepted yield ratio for the proposed method with respect to the Pixie-16 reference system. For each energy point, the accepted timing-event counts within $E \pm 20$ keV were compared using the same coincidence time gate and acquisition live time. This quantity represents a relative ratio between the two methods, rather than an absolute timing-valid efficiency normalized to all triggered events. Horizontal error bars denote the 40-keV-wide integration window, while vertical error bars show the propagated statistical uncertainties. The larger ratio at low energies indicates a substantial recovery of usable timing events in the proposed method compared with the reference processing chain.

The resulting curve shows a clear energy dependence of the improvement. The enhancement is strongest at low energies, reaching more than a factor of two, and progressively decreases as the energy increases. This behavior quantitatively demonstrates that the main advantage of the proposed method lies in its ability to preserve substantially more accepted timing events in the low-energy region, where the conventional Pixie-16 processing suffers the largest loss.

However, a closer comparison between the two matrices (Figs. 6 versus 9) reveals a subtle but significant feature regarding charge collection dynamics. Although the Pixie-16 data appears free of curvature at the bottom, this is an artifact of the threshold cut. Events with the slowest rise times, which are most prone to walking, are simply discarded. In contrast, the 2745 spectrum retains these slow events,

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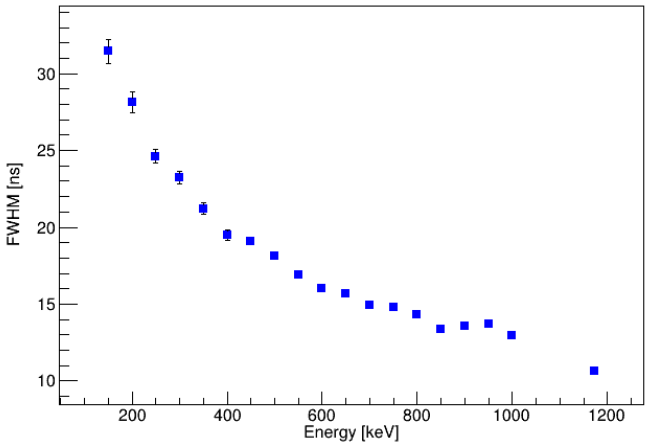


Fig. 12. Timing resolution (FWHM) as a function of gamma-ray energy derived from the proposed algorithm. The system provides continuous timing information down to the low-energy region, limited only by baseline noise.

resulting in a slight, discernible walk feature (curvature) at the lowest energies. Crucially, as these events are preserved rather than rejected, this residual walk remains available for subsequent characterization and offline correction (e.g., quadratic energy-dependent corrections). This capability offers a distinct advantage, which can facilitate even higher timing precision through post-processing without the irreversible loss of statistics inherent to hardware-level thresholding.

Furthermore, the robustness of the algorithm is quantified in Fig. 12, which plots the intrinsic timing resolution (FWHM) as a function of gamma-ray energy. The resolution follows the expected inverse dependence on signal amplitude (approximately 1/E), reflecting the increasing impact of electronic noise on timing jitter at low energies. Even in the challenging low-energy region, the system maintains a stable timing resolution (e.g., ~30 ns at 150 keV), indicating that the validation-threshold-free approach remains effective despite increasing baseline-noise influence.

V. Conclusion

This work addresses the fundamental trade-off between timing precision and detection efficiency in DPP for large-volume HPGe detectors. By implementing a trigger-guided digital CFD architecture without a CFD validation threshold on the CAEN Open FPGA platform, the proposed algorithm decouples the trigger logic from the timing extraction, making it possible to achieve nanosecond-level timing resolution across the energy range investigated in this work. Experimental results with ^{60}Co yielded a time resolution of 10.65 (2) ns, comparable to that of the commercial reference system, while substantially mitigating the low-energy timing-acceptance loss observed in the reference digital CFD implementation.

A key advantage of this architecture is its transparency regarding charge collection dynamics. Compared with the Pixie-16 reference system, the proposed method shows a clear advantage in the low-energy region, where it retains substantially more timing-accepted events. Although this retention reveals a slight residual walk at the low energy, it preserves a substantially larger fraction of the low-energy timing dataset, allowing for further refinement through offline quadratic corrections. The developed firmware provides a comprehensive solution for modern nuclear structure experiments, particularly those relying on gamma-gamma coincidence matrices or lifetime measurements with critical low-energy transitions.

The trigger-guided architecture presented here is not intrinsically limited to CFD. CFD was selected in this work because it offers deterministic latency, modest FPGA resource usage, and a well-established basis for comparison with commercial digital systems. Nevertheless, the same trigger framework can be extended in future work to alternative timing estimators, such as rise time-assisted corrections, waveform fitting around the leading edge, or pulse shape-analysis-based timing reconstruction.

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References

[1] V. T. Jordanov and G. F. Knoll, "Digital synthesis of pulse shapes in real time for high resolution radiation spectroscopy," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 345, no. 2, pp. 337–345, Jun. 1994, doi: (10.1016/0168-9002(94)91011-1.

[2] A. Georgiev and W. Gast, "Digital pulse processing in high resolution, high throughput, gamma-ray spectroscopy," *IEEE Trans. Nucl. Sci.*, vol. 40, no. 4, pp. 770–779, Aug. 1993, doi: 10.1109/23.256659.

[3] H. Tan, M. Momayezi, A. Fallu-Labruyere, Y. X. Chu, and W. K. Warburton, "A fast digital filter algorithm for gamma-ray spectroscopy with double-exponential decaying scintillators," *IEEE Trans. Nucl. Sci.*, vol. 51, no. 4, pp. 1541–1545, Aug. 2004, doi: 10.1109/TNS.2004.832984.

[4] L. Begley et al., "Algorithms of pulse shape analysis for gammasphere under high count rate conditions," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1040, Oct. 2022, Art. no. 167113, doi: 10.1016/j.nima.2022.167113.

[5] H. Y. Wu et al., "The upgrade of the general-purpose digital data acquisition system (GDDAQ)," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 1085, May 2026, Art. no. 171315, doi: 10.1016/j.nima.2026.171315.

[6] J.-P. Martin, C. Mercier, N. Starinski, C. J. Pearson, and P.-A. Amaudruz, "The TIGRESS DAQ/trigger system," *IEEE Trans. Nucl. Sci.*, vol. 55, no. 1, pp. 84–90, Feb. 2008, doi: 10.1109/TNS.2007.910853.

[7] L. C. Mihailescu, C. Borcea, and A. J. M. Plompen, "Data acquisition with a fast digitizer for large volume HPGe detectors," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 578, no. 1, pp. 298–305, Jul. 2007, doi: 10.1016/j.nima.2007.05.231.

[8] F. C. L. Crespi et al., "HPGe detectors timing using pulse shape analysis techniques," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 620, nos. 2–3, pp. 299–304, Aug. 2010, doi: 10.1016/j.nima.2010.02.273.

[9] XIA LLC. (Sep. 13, 2019). *Pixie-16 User Manual*. [Online]. Available: https://www.xia.com/products/pixie-16/

[10] A. Fallu-Labruyere, H. Tan, W. Hennig, and W. K. Warburton, "Time resolution studies using digital constant fraction discrimination," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 579, no. 1, pp. 247–251, Aug. 2007, doi: 10.1016/j.nima.2007.04.048.

[11] H. Y. Wu et al., "A general-purpose digital data acquisition system (GDDAQ) at Peking university," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 975, Sep. 2020, Art. no. 164200, doi: 10.1016/j.nima.2020.164200.

[12] D.-W. Luo et al., "Performance of digital data acquisition system in gamma-ray spectroscopy," *Nucl. Sci. Techn.*, vol. 32, p. 79, Nov. 2021, doi: 10.1007/s41365-021-00917-8.

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